

EEE4114F: Problem set 4 solutions

1. The discrete-time sequence

$$x[n] = \cos\left(\frac{\pi n}{4}\right)$$

results by sampling the continuous-time signal

$$x_c(t) = \cos(\Omega_o t).$$

Since $\omega = \Omega T$ and $T = 1/1000$ seconds, the signal frequency could be:

$$\Omega_o = \frac{\pi}{4} \cdot 1000 = 250\pi$$

or possibly:

$$\Omega_o = \left(2\pi + \frac{\pi}{4}\right) \cdot 1000 = 2250\pi.$$

2. (a) Letting $T = 1/100$ gives

$$\begin{aligned} x[n] &= x_c(nT) \\ &= \sin\left(20\pi n \frac{1}{100}\right) + \cos\left(40\pi n \frac{1}{100}\right) \\ &= \sin\left(\frac{\pi n}{5}\right) + \cos\left(\frac{2\pi n}{5}\right) \end{aligned}$$

- (b) No, another choice is $T = 11/100$:

$$\begin{aligned} x[n] &= x_c(nT) \\ &= \sin\left(20\pi n \frac{11}{100}\right) + \cos\left(40\pi n \frac{11}{100}\right) \\ &= \sin\left(\frac{11\pi n}{5}\right) + \cos\left(\frac{22\pi n}{5}\right) \\ &= \sin\left(\frac{\pi n}{5}\right) + \cos\left(\frac{2\pi n}{5}\right) \end{aligned}$$

3. The continuous-time signal contains an attenuated replica of the original signal with a delay of τ_d .

$$x_c(t) = s_c(t) + \alpha s_c(t - \tau_d)$$

- (a) Taking the Fourier transform of the analog signal:

$$X_c(j\Omega) = S_c(j\Omega) \cdot (1 + \alpha e^{-j\tau_d\Omega})$$

Note that $X_c(j\Omega)$ is zero for $|\Omega| > \pi/T$. Sampling the continuous-time signal yields the discrete-time sequence, $x[n]$. The Fourier transform of the sequence is

$$\begin{aligned} X(e^{j\omega}) &= \frac{1}{T} \sum_{r=-\infty}^{\infty} S_c\left(\frac{j\omega}{T} + j\frac{2\pi r}{T}\right) \\ &\quad + \frac{\alpha}{T} \sum_{r=-\infty}^{\infty} S_c\left(\frac{j\omega}{T} + j\frac{2\pi r}{T}\right) e^{-j\tau_d\left(\frac{\omega}{T} + \frac{2\pi r}{T}\right)}. \end{aligned}$$

- (b) The desired response:

$$H(j\Omega) = \begin{cases} 1 + \alpha e^{-j\tau_d\Omega}, & \text{for } |\Omega| \leq \frac{\pi}{T} \\ 0, & \text{otherwise} \end{cases}$$

Using $\omega = \Omega T$, we obtain a discrete-time system which simulates the above response:

$$H(e^{j\omega}) = 1 + \alpha e^{-j\frac{\tau_d\omega}{T}}$$

- (c) We need to take the inverse Fourier transform of the discrete-time impulse response of part (b).

$$\begin{aligned} h[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} H(e^{j\omega}) e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} (1 + \alpha e^{-j\frac{\tau_d\omega}{T}}) e^{j\omega n} d\omega \end{aligned}$$

- (i) Consider the case when $\tau_d = T$:

$$\begin{aligned} h[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} (e^{j\omega n} + \alpha e^{j\omega(n-1)}) d\omega \\ &= \frac{\sin(\pi n)}{\pi n} + \frac{\alpha \sin[\pi(n-1)]}{\pi(n-1)} \\ &= \delta[n] + \alpha \delta[n-1] \end{aligned}$$

- (ii) For $\tau_d = T/2$:

$$\begin{aligned} h[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} (e^{j\omega n} + \alpha e^{j\omega(n-\frac{1}{2})}) d\omega \\ &= \frac{\sin(\pi n)}{\pi n} + \frac{\alpha \sin[\pi(n-\frac{1}{2})]}{\pi(n-\frac{1}{2})} \\ &= \delta[n] + \frac{\alpha \sin[\pi(n-\frac{1}{2})]}{\pi(n-\frac{1}{2})} \end{aligned}$$

4. (a) Notice first that $H(e^{j\omega}) = 10j\omega$, $-\pi \leq \omega < \pi$.

(i) After sampling,

$$\begin{aligned} x[n] &= \cos\left(\frac{3\pi}{5}n\right), \\ y[n] &= |H(e^{j\frac{3\pi}{5}})| \cos\left(\frac{3\pi}{5}n + \angle H(e^{j\frac{3\pi}{5}})\right) \\ &= 6\pi \cos\left(\frac{3\pi}{5}n + \frac{\pi}{2}\right) \\ &= -6\pi \sin\left(\frac{3\pi}{5}n\right) \\ y_c(t) &= -6\pi \sin(6\pi t). \end{aligned}$$

(ii) After sampling, $x[n] = \cos(\frac{7\pi}{5}n) = \cos(\frac{3\pi}{5}n)$, so again, $y_c(t) = -6\pi \sin(6\pi t)$.

(b) $y_c(t)$ is what you would expect from a differentiator in the first case but not in the second case. This is because aliasing has occurred in the second case.

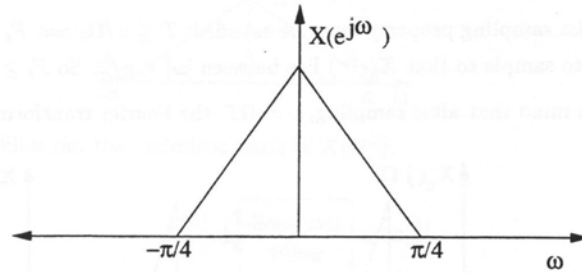
5. There is no loss of information if $X(e^{j\omega/2})$ and $X(e^{j(\omega/2-\pi)})$ do not overlap. This is true for (b), (d), (e).

6. The output $x_r[n] = x[n]$ if no aliasing occurs as result of downsampling. That is, $X(e^{j\omega}) = 0$ for $\pi/3 \leq |\omega| \leq \pi$.

(a) $x[n] = \cos(\pi n/4)$. $X(e^{j\omega})$ has impulses at $\omega = \pm\pi/4$, so there is no aliasing. $x_r[n] = x[n]$.

(b) $x[n] = \cos(\pi n/2)$. $X(e^{j\omega})$ has impulses at $\omega = \pm\pi/2$, so there is aliasing. $x_r[n] \neq x[n]$.

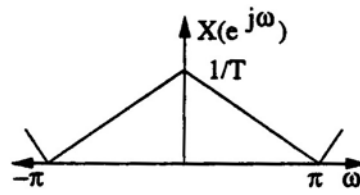
(c) A sketch of $X(e^{j\omega})$ is shown below. Clearly there will be no aliasing and $x_r[n] = x[n]$.



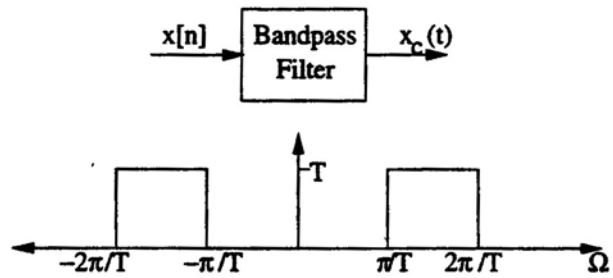
7. The nyquist sampling property must be satisfied: $T \leq \pi/\Omega_0$.

8. (a)

$$\omega = \Omega T, \quad T = \frac{2\pi}{\Omega_0}$$



(b) To recover simply filter out the undesired parts of $X(e^{j\omega})$.



(c)

$$T \leq \frac{2\pi}{\Omega_0}$$

9. The Fourier transform of $y_c(t)$ is sketched below for each case.

